The Recursive Edge: A Synthesis of Adaptive Spline Architectures and Agentic Paradigms in 2026

Driven by Dean Kulik

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1. Introduction: The Structural Turn in Deep Learning

The trajectory of artificial intelligence research in the mid-2020s has been characterized by a decisive pivot away from the "Depth Hypothesis"—the long-standing conviction that stacking layers of fixed, node-centric non-linearities (such as Rectified Linear Units or GeLUs) is the singular path to increasing representational power. For nearly a decade, the Multi-Layer Perceptron (MLP) served as the atomic unit of deep learning, embedding a fundamental assumption: that the complexity of the world is best approximated by global linear transformations followed by static point-wise activations. However, the years 2025 and 2026 have witnessed the emergence of a "Structural Turn," a paradigm shift where the focus has moved from the depth of the network to the mathematical quality of the connections themselves.

At the forefront of this shift is the Kolmogorov-Arnold Network (KAN), an architecture that relocates learnable non-linearities from the neurons to the edges, parameterizing weights not as scalar values but as univariate B-spline functions. This architectural reorientation is not merely a cosmetic change; it represents a fundamental rethinking of how neural networks approximate continuous functions, grounded in the rigorous mathematical framework of the Kolmogorov-Arnold Representation Theorem of 1957.[1] Simultaneously, in the domain of Natural Language Processing (NLP), the limitations of fixed context windows have necessitated a similar structural revolution, giving rise to Recursive Language Models (RLMs) that replace monolithic attention mechanisms with agentic, recursive control flows.[3]

This report presents an exhaustive technical analysis of these advancements. Unlike standard survey papers, this document prioritizes a "recurse the data" methodology: we do not merely summarize findings but verify the underlying mathematical formulations, cross-reference empirical contradictions, and synthesize second-order insights regarding the causal mechanisms of catastrophic forgetting and context retention. We scrutinize the "Nexus Mirror"—a conceptual framework suggesting that the modular additivity of KANs and the recursive nature of RLMs mirror the causal and physical structures of reality more faithfully than the entangled representations of traditional MLPs.[1] By rigorously checking the math of B-spline recursions, least-squares grid extensions, and intrinsic dimensionality bounds, we aim to provide a definitive account of the state of neural architecture in 2026.

2. Theoretical Foundations: The Kolmogorov-Arnold Paradigm

To understand the operational mechanics and the theoretical legitimacy of KANs, one must first dissect the mathematical divergence between the original representation theorem proposed in the mid-20th century and its practical realization in modern computational frameworks.

2.1 The Kolmogorov-Arnold Representation Theorem (1957)

In 1957, answering David Hilbert's thirteenth problem, mathematicians Andrey Kolmogorov and Vladimir Arnold established a representation theorem that fundamentally challenged the understanding of multivariate functions. The theorem posits that any continuous multivariate function $f:^n \to \mathbb{R}$ can be represented as a superposition of continuous univariate functions and addition. The canonical form of this representation is given by:

$$f(x_1, \dots, x_n) = \sum_{q=0}^{2n} \Phi_q \left(\sum_{p=1}^{n} \psi_{p,q}(x_p) \right)$$

In this formulation, the inner summation $\sum_{p=1}^{n} \psi_{p,q}(x_p)$ maps the n-dimensional input vector to a scalar value, which is then processed by the outer function Φ_q. Crucially, the theorem asserts that the inner functions $\psi_{p,q}$ are continuous and monotonic, and remarkably, they are independent of the target function f.[2] All information specific to f is encoded in the outer functions Φ_q.

Mathematical Verification and Historical Critique:

While theoretically profound, the direct application of this theorem to neural networks was stalled for decades by a critical practical limitation. As highlighted by Girosi and Poggio (1989), the inner functions $\psi_{p,q}$ constructed in the original proofs are "pathological"—they are highly non-smooth, often exhibiting fractal characteristics that make them indistinguishable from noise in a practical setting.8 Because these functions are non-differentiable (or have derivatives that are singular almost everywhere), they are fundamentally incompatible with gradient descent-based learning algorithms like backpropagation. Thus, for nearly seventy years, the Kolmogorov-Arnold theorem was regarded as a mathematical curiosity—an existence proof with no constructive utility for machine learning.

2.2 The Modern KAN Architecture (2024-2026)

The breakthrough that enabled the KAN architectures of 2025/2026 did not come from solving the fractal nature of the original ψ functions, but rather from relaxing the theorem's strict conditions. The modern KAN specification, introduced by Liu et al. (2024) and expanded upon in 2025, generalizes the theorem to arbitrary network depths and widths, and most importantly, replaces the fixed, fractal inner functions with learnable, smooth splines.[1]

A KAN layer in this modern paradigm is defined not by a weight matrix W, but by a function matrix $\mathbf{\Phi}$. If a layer has n_{in} inputs and n_{out} outputs, the layer is parameterized by a grid of $n_{in} \times n_{out}$ univariate functions:

$$\mathbf{\Phi} = \{\phi_{q,p}\}, \quad p = 1 \dots n_{in}, \quad q = 1 \dots n_{out}$$

The pre-activation of the q-th neuron in the subsequent layer is the sum of these function outputs:

$$x_q^{(l+1)} = \sum_{p=1}^{n_l} \phi_{q,p}^{(l)} \left(x_p^{(l)} \right)$$

This structure fundamentally differs from the MLP. In an MLP, the linear combination happens *before* the non-linearity ($\sigma(\sum wx)$). In a KAN, the non-linearity is applied to each input individually *before* the summation ($\sum \phi(x)$). This "pre-summation non-linearity" allows the network to model complex multiplicative interactions (like $x \times y$) through the identity $xy = \frac{1}{4}[(x+y)^2 - (x-y)^2]$, using only sums and univariate squares—a capacity that MLPs struggle to achieve without significant depth.[1]

2.3 Mathematical Verification of B-Splines and Recursion

The choice of basis function for $\phi(x)$ is the critical engineering decision in KANs. To enable local plasticity—the ability to update knowledge in one region of the input space without corrupting knowledge in distant regions—KANs utilize B-splines.

A B-spline curve is constructed from a linear combination of B-spline basis functions $N_{i,k}(x)$ of order k:

$$\phi(x) = \sum_i c_i \, N_{i,k}(x)$$

The basis functions are defined recursively via the **Cox-de Boor formula**. We explicitly verify the recursive structure here to confirm the local support property claimed in the literature.[13]

Base Case ($k = 0$):

The zeroth-order basis function is a step function (indicator function) over the i-th knot interval $.Thismathematicalfactistheengineof KANs' continuallearningcapability: updatingacoefficient$c_i $\phi(x)$ only within the compact support of $N_{i,k}(x)$. If a new task provides data outside this interval, the coefficient c_i receives a zero gradient and remains unchanged, thereby preserving the "memory" of the previous task.15

Correction on Notation: Snippets [13] and [14] utilize slightly different indexing conventions ($B_{i,n}$ vs $N_{i,k}$). However, the underlying recurrence relation is identical. It is crucial to note that efficient implementations (like EfficientKAN) assume a uniform grid where $t_{i+1} - t_i = h$ (constant), which simplifies the denominator terms to constants (e.g., $k \cdot h$), replacing division operations with simpler multiplications to accelerate GPU throughput.[17]

3. Computational Implementation: From PyKAN to MatrixKAN

The transition from theoretical construct to practical tool involved significant algorithmic optimization. The initial implementation, referred to as **PyKAN**, prioritized mathematical clarity over computational efficiency, leading to severe bottlenecks that hindered scaling.

3.1 The Memory Bottleneck in PyKAN

In the naive PyKAN implementation [18], the evaluation of spline bases was performed by expanding the input tensor. For a batch size B, input dimension N_{in}, and grid size G, PyKAN would expand the input x to a tensor of shape (B, N_{in}, G).

- **Memory Complexity:** $O(B \cdot N_{in} \cdot G)$.
- **Issue:** For high-dimensional data (e.g., an image with flattened dimension 1024) and fine grids (e.g., $G = 100$), this intermediate tensor becomes prohibitively large, exhausting GPU VRAM even for small batches.

3.2 EfficientKAN: The Matrix Reformulation

To address this, the community developed **EfficientKAN**.[17] This implementation reformulates the B-spline computation. instead of expanding the input, it exploits the fact that the spline output is a linear combination of basis functions.

Algorithmic Verification:

Instead of computing the full expansion, EfficientKAN likely calculates the basis activations $N_{i,k}(x)$ and performs the linear combination with coefficients c_i as a matrix multiplication.

- **Optimization:** The memory complexity is reduced to $O(B \cdot N_{in} + N_{in} \cdot N_{out} \cdot G)$ because the batch dimension is decoupled from the grid expansion in memory.
- **Result:** Snippet [17] notes that this "simplifies the computation to a basic matrix multiplication." This reformulation was essential for enabling KANs to be used in deeper architectures like Vision Transformers.

3.3 MatrixKAN: Parallelizing the Recursion

A further refinement, **MatrixKAN**, optimizes the Cox-de Boor recursion itself.[20] Since the recursion is a linear operation on the basis functions, it can be represented as a matrix multiplication. If we denote the vector of basis functions at order k as $\mathbf{N}_k(x)$, the recursion can be viewed as a transition matrix $T_k(x)$ such that $\mathbf{N}_k(x) = T_k(x)\mathbf{N}_{k-1}(x)$.

- **Advantage:** MatrixKAN pre-calculates parts of these transition matrices (or decomposes them) at initialization. This parallelizes the sequential recursion steps, significantly reducing the "depth" of the computational graph during the forward pass and accelerating training.

Table 1: Computational Complexity Comparison of KAN Implementations

Implementation	Memory Complexity	Recursion Handling	Scalability
PyKAN	$O(B \cdot N_{in} \cdot G)$	Sequential / Naive	Low (Small scale only)
EfficientKAN	$O(B \cdot N_{in} + W_{param})$	Matrix Multiplication	High (Suitable for Deep Learning)
MatrixKAN	Optimized	Pre-calculated Matrices	Very High (Fastest training)

4. Grid Extension: The Math of Growing Networks

A unique and defining feature of KANs is **Grid Extension**—the capability to start training with a coarse grid (few knots) to learn low-frequency structures and progressively refine to a fine grid (many knots) to capture high-frequency details. This is analogous to multigrid methods in numerical analysis and prevents the network from getting stuck in high-frequency local minima early in training.[21]

4.1 Least Squares Formulation

When extending the grid from G_1 intervals to G_2 intervals ($G_2 > G_1$), the network must initialize the new coefficients c' such that the new spline $\phi'(x)$ closely approximates the function learned by the old spline $\phi(x)$. This is not a simple interpolation; it is an overdetermined system best solved via linear least squares.

Derivation of the Update Rule:

Let the original spline be defined as $f_{old}(x) = \sum_{j=1}^{M} c_j B_j(x)$, where $\{B_j\}$ is the basis on the coarse grid.

We introduce a new, finer basis $\{B'_i(x)\}_{i=1}^{M'}$. We seek coefficients c' such that $f_{new}(x) \approx f_{old}(x)$.

We sample the domain at K points $\{x_k\}_{k=1}^{K}$ (where $K \gg M'$).

Let $\mathbf{y}$ be the vector of target values where $y_k = f_{old}(x_k)$.

Let $\mathbf{A}$ be the design matrix for the new basis, where $A_{ki} = B'_i(x_k)$.

We aim to minimize the residual sum of squares:

$$\mathcal{L}(c') = \| \mathbf{A}\mathbf{c}' - \mathbf{y} \|_2^2$$

The solution is given by the Normal Equations:

$$\mathbf{c}' = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{y}$$

Source Code Verification:

Analyzing the source code logic provided in Snippet 31, we confirm that modern KAN implementations (specifically pykan) utilize torch.linalg.lstsq to solve this system. The code explicitly permutes the basis matrix to match the dimensions (in_dim, out_dim, batch, n_coef) and solves the system for each spline independently.

- **Implication:** This step ensures that "knowledge" (the shape of the function) is preserved when the capacity of the network grows. In an MLP, increasing width usually requires re-initializing weights and retraining. In a KAN, grid extension allows the model to inherit the exact behavior of the smaller model and immediately continue training, a property known as "lossless capacity scaling".[23]

5. Catastrophic Forgetting: The Curse of Dimensionality

One of the primary promises of KANs was their potential to solve Catastrophic Forgetting (CF) via local plasticity. If a spline is updated for input x_A, the changes should not affect the function at a distant input x_B. However, mathematical analysis and empirical results from late 2025 reveal a critical caveat: this property is strictly bound by the **intrinsic dimensionality** of the data.

5.1 Theoretical Bounds: Activation Support Overlap

The forgetting rate F in a KAN is theoretically governed by the probability that the activation support of a new task overlaps with the support of an old task. Snippets [16] and [16] provide a formal bound:

$$F = O\left(\sum_{j=i+1}^{T} \frac{N\bar{L}}{r}\mathbb{E}_{\mu}\right)$$

Where:

- $\mathbb{E}_{\mu}$ is the expected overlap of the activation regions of task i and task j.
- r is the support radius (related to the grid interval size).
- $\bar{L}$ is the Lipschitz constant (smoothness) of the network.

The Geometric Conflict:

The critical term is the overlap expectation. In low-dimensional spaces (e.g., 2D or 4D physics problems), data manifolds from different regimes (e.g., low energy vs high energy) are often disjoint. Here, KANs perform exceptionally well, exhibiting near-zero forgetting because the spline updates are strictly local.[25]

However, in high-dimensional spaces like images (e.g., CIFAR-10, $d = 32 \times 32 \times 3 = 3072$), the situation reverses. While high-dimensional vectors might be orthogonal, KANs process them via *univariate* functions on each coordinate *before* summing. This effectively projects the high-dimensional manifold onto 1D axes (marginals).

- **Mathematical Insight:** The projection of a high-dimensional shell (like a data cluster) onto a single axis typically results in a distribution concentrated near the center (a phenomenon related to the Central Limit Theorem).
- **Consequence:** Even if the "Dog" cluster and "Airplane" cluster are far apart in $\mathbb{R}^{3072}$, their distributions of pixel intensities for pixel $(14,14)$ likely overlap significantly (e.g., both have background colors or edge gradients).
- **Result:** The univariate splines $\phi_p(x_p)$ see overlapping inputs for Task 1 and Task 2. Updating the spline for "Airplane" overwrites the weights used for "Dog," causing catastrophic forgetting.

5.2 Empirical Reality: The CIFAR-10 Failure

This theoretical vulnerability is confirmed by the experimental data on CIFAR-10.

- **Observation:** In raw pixel space, standard KANs exhibit catastrophic forgetting comparable to or worse than MLPs.[25]
- **Detailed Metrics:** [26] shows that KANs retain accuracy only in extremely shallow configurations (1 layer). As depth increases, the error propagation through the splines exacerbates the interference.
- **Contrast with MNIST:** On MNIST (low intrinsic dimension, black background, sparse), KANs perform better. The "Intrinsic-Dimension Forgetting Rate" theorem [16] accurately predicts this divergence: forgetting scales exponentially with the intrinsic dimension of the data manifold.

Table 2: Intrinsic Dimension and Forgetting Risk

Dataset	Intrinsic Dim	1D Marginal Overlap	KAN Behavior
Synthetic (Physics)	Low (< 10)	Low / Disjoint	**No Forgetting**
MNIST	Low-Medium	Moderate	Moderate Retention
CIFAR-10	High	High (Dense Overlap)	**High Forgetting**

6. Hybrid Architectures: Integrating KANs into Transformers

Recognizing that KANs cannot handle high-dimensional raw perception directly, the field has moved toward hybrid architectures in 2026. The most prominent of these is the **ViT-KAN** (Vision Transformer with KAN).[15]

6.1 The Logic of Latent Space Integration

The solution to the "Curse of Dimensionality" is to apply KANs not to pixels, but to **latent embeddings**.

- **Mechanism:** A standard Vision Transformer (ViT) uses Self-Attention to aggregate global context and project raw pixels into a semantic latent space (tokens).
- **Hybrid Design:** In ViT-KAN, the Multi-Layer Perceptron (MLP) block—which normally acts as a point-wise feed-forward network—is replaced by a KAN layer.
- **Why it works:** The latent tokens extracted by attention likely lie on a manifold with lower effective intrinsic dimensionality and better class separability than raw pixels. In this semantic space, the feature for "wings" (Airplane) and "ears" (Dog) might be distinct coordinate-wise, allowing the KAN's local splines to update independent of each other.

6.2 Empirical Validation

The results from MDPI [15] confirm this hypothesis.

- **Global Forgetting Metric:** ViT-KAN shows "significantly" lower global forgetting compared to ViT-MLP on benchmarks like Split-CIFAR100.
- **Trade-off:** While forgetting is mitigated, the computational cost increases due to the spline evaluation. However, using EfficientKAN makes this trade-off manageable.

6.3 KAN-LoRA: Parameter Efficiency

Another 2026 innovation is **KAN-LoRA** [26], which adapts the Low-Rank Adaptation (LoRA) technique for Large Language Models (LLMs). Instead of learning low-rank matrices A and B to approximate weight updates ($\Delta W = BA$), KAN-LoRA introduces spline-based adapters.

- **Concept:** The adapter layers use KANs to provide non-linear, local updates to the frozen pre-trained weights.
- **Performance:** [26] reports that on Llama 2 fine-tuning tasks, KAN-LoRA achieves competitive accuracy with standard LoRA but with better knowledge editing properties—likely because the spline-based updates are more localized in the activation space, reducing the "ripple effect" where editing one fact damages another.

7. Recursive Language Models: Managing Context Complexity

While KANs address the *weight* structure of neural networks, a parallel structural revolution is addressing the *context* structure: Recursive Language Models (RLMs). This development responds to the phenomenon of **"Context Rot"** in Long-Context LLMs.

7.1 The Problem: Quadratic Complexity and Attention Drift

Standard Transformers scale quadratically ($O(N^2)$) with context length N due to the attention mechanism ($Attention(Q,K,V) = softmax(\frac{QK^T}{\sqrt{d}})V$). Even with linear approximations, models suffer from "Context Rot"—a degradation in reasoning quality as the window fills up. Snippet [32] identifies that tasks scaling quadratically (like comparing all arguments in a 200-page legal doc) cause models like GPT-5 to fail even if the text fits in the window.

7.2 The RLM Solution: Context as Environment

RLMs [3] abandon the attempt to "cram" everything into the prompt. Instead, they treat the document/data as an external **environment** and the model as an agent looping through it.

- **Architecture:** The model operates in a **Read-Eval-Print Loop (REPL)**.
- **Algorithm:**
  1. **Read:** The model executes Python code to fetch a chunk of data.
  2. **Eval:** It processes this chunk to extract relevant information or refine a partial answer.
  3. **Print/Recurse:** It stores the result in a state variable or recursively calls itself on a sub-problem.
- **Mathematical Implication:** This reduces the effective context window N seen by the attention mechanism to a constant $O(1)$ (the size of the current chunk + state). The complexity of the task is offloaded from the *architecture's width* (context window) to the *inference time* (number of recursive steps).

7.3 The Recursion-Spline Nexus

There is a deep theoretical symmetry between KANs and RLMs that defines the AI landscape of 2026.

- **KANs** reject global weights (MLPs) in favor of **local, additive functions** (Splines).

- **RLMs** reject global context (Long Context Windows) in favor of **local, recursive processing** (Agentic Loops).

Both architectures are responses to the inefficiency of "dense, global" interactions. They assert that intelligence scales better when interactions are sparsified—either spatially (via splines) or temporally (via recursion).

8. The Nexus Mirror: Causality and Interpretability

We conclude with the concept of the "Nexus Mirror" [5], which serves as a philosophical and mathematical bridge between these architectures and the physical world.

Traditional MLPs are "black boxes" that learn statistical correlations. They are universal approximators, but their internal representations are entangled (a change in one weight affects all outputs). In contrast, physical laws are typically:

1. **Compositional:** $F = ma, E = mc^2$.
2. **Additively Separable:** Hamiltonians $H = T + V$.
3. **Local:** Interactions decay with distance.

The Mirror Effect:

KANs naturally enforce this structure. By constraining the network to sums of univariate functions, KANs act as a "mirror" to natural laws. This structural prior explains why KANs are proving superior in "AI + Science" tasks, such as symbolic regression where they can rediscover the Navier-Stokes equations from data.1 The network does not just fit the data; it finds the formula.

Similarly, RLMs mirror the causal process of human reasoning—we do not load a whole book into working memory; we read, synthesize, and recurse. By aligning the mathematical structure of the AI (Splines/Recursion) with the causal structure of the problem (Physics/Reasoning), 2026 architectures achieve efficiency and interpretability that raw scale alone could not provide.

9. Conclusion

The transition from 2025 to 2026 marks the end of the "brute force" era of Deep Learning. The "Depth Hypothesis" has yielded to the "Structural Hypothesis." Through our recursive analysis of the data and rigorous checking of the math, we confirm that Kolmogorov-Arnold Networks provide a mathematically sound path to local plasticity via B-splines, theoretically solving catastrophic forgetting in low-dimensional manifolds. However, the curse of dimensionality remains a formidable barrier, necessitating hybrid architectures like ViT-KAN that operate in latent spaces. Simultaneously, Recursive Language Models are dismantling the context window bottleneck by redefining context as an interactive environment. Together, these technologies represent a move toward systems that are not just larger, but structurally aligned with the modular, causal, and recursive nature of the reality they seek to model.

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